



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION - SEMESTER I (2020/2021)
CSC1041 - PROGRAMMING LABORATORY I

Answer **ALL** Questions

Time Allowed: **Three** Hours

PART A - (C PROGRAMMING)

Save your C program as S19XXXQ1.c where S19XXX is your registration number. Extra marks will be awarded for using good programming practices.

1. [50 Marks]

Create a C console application for an Expressway-Bus ticket purchasing system. A user should be able to buy a maximum of 3 seats for a single destination.

- a) Create separate functions to print start interchanges and destination interchanges. Expressway prices are given below. (Hint: you may use an appropriate data structure to store below ticket prices)

start \ end	Kadawatha	Kaduwela	Athurugiriya	Kottawa
Kaduwela	100			
Athuruguruya	150	100		
Kottawa	150	100	100	
Kahathuduwa	200	150	150	100

Each seat has an additional bus charge of Rs.50.00

- b) A recurring menu should have an option to buy(book) tickets. It should ask to input the passenger's Name, NIC number, Mobile number, and ticket details. You are required to have an appropriate mechanism to store all the ticket bookings in the system.
- c) The menu should have an option to search a passenger by his/her NIC number. The appearance of the main menu is given below.

Example output:

```
Southern Expressway ticket booking
***** select an option from the menu. *****
1.Buy tickets.
2.Search Passenger by ID.
9.Exit.
option:
```

After booking a ticket, the purchase summary should be printed as below.

Example output:

....Booking added successfully....

Name : Kamal
ID NO : 890456123v
Mobile : 078199347
Departure : Kottawa
Destination : Kadawatha
Seats : 3
Total (Rs) : 600.00

PART B - (PYTHON PROGRAMMING)

Create a folder named CSC1041_End_S19XXX in your H drive. Copy the shared text files into the newly created folder. Save your Python programs inside the same folder as S19XXXQ2.py, S19XXXQ3.py, S19XXXQ4.py for each question where S19XXX is your registration number.

2. [30 Marks]

A. File encryption is a way of encoding files, including the sensitive data they contain, in order to transfer them securely. Encoding prevents unauthorized access and tampering by malicious actors. It prevents a file from being read by anyone except the person for whom it was intended. You are required to design a program which has the ability to encrypt and decrypt text files. You may consider the instructions below, use appropriate collection data types and create necessary functions to generate the final output text files (normal text file encrypted.txt, encrypted text file decrypted.txt). For the encryption process first convert all the letters into lowercase letters and then consider the following. The letter which is being processed should be converted (encoded) as follows.

i. If the character (letter) is a vowel,

a → Q	e → W	i → E	o → R	u → T
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ii. If the character (letter) contains a numeric value (from 0 to 9),

The numeric value should be shifted by +4			
0 → 4	1 → 5	6 → 0	8 → 2

iii. If the character (letter) is a symbol or a letter below,

% → A	, → S	& → D	(→ L	) → K	! → @	y →	x → {
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iv. If the character (letter) is a consonant (letter other than vowels), the character ASCII value should be shifted by +3. [Hint : Use **chr()** and **ord()** in-built functions.]

chr() - Converts ASCII-based number to character

chr(65) --> 'A'

ord() - returns an integer of the given single Unicode character

ord('A') --> 65

character ASCII value should be shifted by +3			
a → d	e → h	t → w	m → p

v. If the character (letter) is a dash(-) or a blank space (" ")

“ ” → “_”	“-” → “ ”
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B. Create necessary functions and data structures to decrypt a given text file. You need to use the above same encryption procedure in reverse.

C. Read the given text file ('normal_text_file.txt') and encrypt the whole in it and write encrypted text into 'normal_text_file_encrypted.txt' file

D. Read the given text file ('encrypted_text_file.txt') and decrypt the whole content and write decrypted text into 'encrypted_text_file_decrypted.txt' file

Example:

When using the encryption process below text should be encrypted as follows

Hello world 1900! → kWooR-zRuog-5344@

When using the decryption process below text should be decrypted as follows

kWooR-vTppWu-D-5399@ → hello Summer & 1955!

kWooR-zRuog-5344@ → hello world 1900!

3. [10 Marks]

Body mass index (BMI) is an inexpensive and easy screening method for understanding the weight category of a particular person. You are required to create a function to calculate the BMI from the given height and weight of a person. Then copy the given dictionary (persons) with exact values and symbols into your python script and print the given example output using the already created dictionary and the BMI calculator function.

$$BMI = \frac{Weight}{Height * Height}$$

```
persons = {'John':{'height':'1.51m', 'weight':'45Kg'}, 'Mike':{'height':'1.90m', 'weight':'95Kg'},  
          'Locke':{'height':'1.76m', 'weight':'78Kg'}, 'Scott':{'height':'1.87m', 'weight':'46Kg'},  
          'Julia':{'height':'1.55m', 'weight':'54Kg'}}
```

Example output:

*****BMI*****

```
-----  
Person height weight BMI  
Peter 1.51 45 19.74  
Mike 1.90 95 26.32  
Locke 1.76 78 25.18  
Scott 1.87 46 13.15  
Julia 1.55 54 22.48
```

4. [10 Marks]

A. Create a Python function [*replace_letters()*] that takes two single strings as user inputs and **swaps** the last letter of each word. In addition to that, the program should consider the following.

- It should remove leading and trailing spaces for both strings
- If the length of the first word is 5, you should replace the first letter of the first word with '@' symbol
- If the total length of the final string is 10 (without considering blank spaces), you should replace the last letter of the final string with '&' symbol.

Example output:

Ex:

Enter the word 1: hello

Enter the text 2: world

Output : @e1ld worl&

Additional Examples

'read', 'write' → "reae writd"

'google', 'services' → "googls servicee"

'final', 'destiation' → "@inan destiat1ol"